HaluMem: Evaluating Hallucinations in Memory Systems of Agents

- Memory systems are important in the current LLM agent systems. However, it is often seen that such memory systems can introduce hallucination, namely "Memory Hallucination", which will decrease the quality of material operations and usage. HaluMem is a framework that will evaluate hallucination-related performance of memory systems.
- Problem Definitions:
  - Let Sbe the memory system in an agent system A.
  - Let $\{(u_i, a_i)\}|_i$ be the interaction set of the user $u$and the agent a.
  - Each memory point will be saved as text, denoted as m.
  - Operations $\{E, U, R, Q\}$, where
    - E: extract memory from conversation
    - U: update of the memory: merging, deletion...
    - R: memory retrival
    - Q: memory-related question answering: construct prompt+model inference
  - End2end accuracy: $Acc_{e2e} = \frac{1}{J}\sum_{i=1}^{J}\mathbb{I}[\hat{y}_i = y_i^*]$. However, this function will not be able to track the specific point of incorrectness (during which operation?).
  - To identify the finer level of the source of mistakes, several *gold standards* are introduced.
    - Extraction gold standards: $G^{ext} = \{m_i\}|_{i=1}^{K}$, represents the *memory points* that *should be* added during the dialogue.
    - Update gold standards: $G^{upd} = \{m^{old} \rightarrow m^{new}\}$representing the set of *memory point* pairs before and after updates during the dialogue.
    - Question-answer dataset: for each question q_i, a gold answer $y_i^*$is provided

      $\hat{M} = E(D),\ \hat{G}^{upd} = U(\hat{M}, D),\ \hat{y}_j = A(R(\hat{M}, q_j), q_j)$, where $\hat{M}^{ext}$ means the set of memory points representing the current state of the memory system when query $q_j$is processed.
- Constructing HaluMem: 6+1 steps
  1. Persona Creation
     - Includes Core Profile Information, Dynamic State Information, and Preference Information.
  2. Life Skeleton
     - Defines evolutionary trajectory
     - Defines an anchor event, which will be the trigger of the evolution of dynamic event
     - Defines preference evolution
  3. Event Flow
     - Includes the flow of events. Each event has attributes including name, type (init events, career events, daily events), time, and description.
  4. Session Summaries and Memory Points

    - By simulating "open-book" conversation, generating memory points and summary.
  5. Session Generation
    - Simulate multiple-round conversation with adversarial information by using the memory points generated by step d.
  6. Question Generation
    - With the multiple-round conversation from step e and memory points from session d, construct final question-answer pairs, where the answer here is used as "golden pairs".
  7. Human Annotation

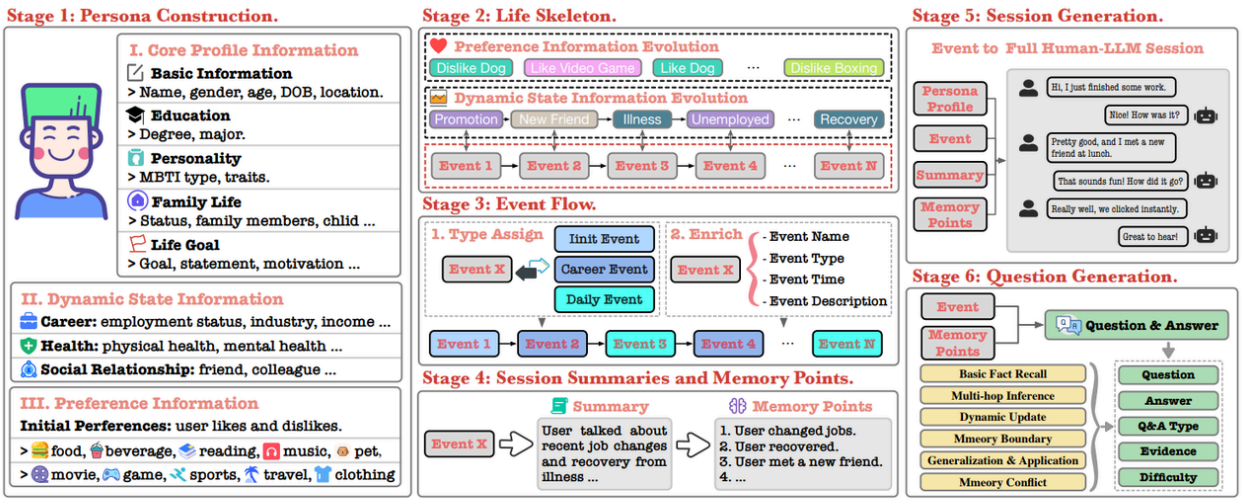

Figure 3 Framework of the HaluMem Construction Pipeline.

- Eval Framework
  - Steps:
    - Dialogue set: $\{D^i\}|_{i=1}^{j}$will be fed to memory system Schronologically.
    - If current Dcontains reference memory points or QA tasks, the corresponding evaluation process (extraction, updating, or question answering) is triggered immediately after S completes processing that session, and the results are recorded.
    - Afterwards, all three metrics will be aggregated together.
  - Auxillary APIs: `Add Dialogue`, `Dialogue Memory`, `Retrieve Memory`
- Metrics: *Memory Extraction, Memory Updating, and Memory Question Answering*.
  - Memory Extraction
    - For *Anti-amnesia*, we have the following matrices (it's all about what it should remember and what it actually remembers):
      - (For D^s) System provides gold standards $G_s^{ext} = \{m_i^s\}|_{i=1}^{K_s}$. The framework will output $\hat{M}_s^{ext} = \{\hat{m}_i^s\}|_{i=1}^{\hat{K}_s}$.
      - Memory Recall = $\frac{N_{correct}}{|G_s^{ext}|}$, where $N_{correct}$is the # of correct retrivals in $\hat{M}_s^{ext}$.
      - Weighted Memory Recall = $\frac{\sum_{i=1}^{|G_s^{ext}|} w_i s_i}{\sum_{i=1}^{|G_s^{ext}|} w_i}$, where w represents the importance weight of the i-th memory, and $s_i$is the retrieval quality score, $s_i \in \{0, 0.5, 1\}$(from the worst to the best).
    - For *Anti-hallucination*, meaning for all retrieved memory, formally $\hat{M}_s^{ext}$, we check the

number of **correctly retrieved** memory.

- Extract Acc = $\frac{\sum_{i=1}^{|\hat{M}_s^{ext}|} s_i}{|\hat{M}_s^{ext}|}$.
- Extract precision, meaning for all retrieved memories that are in the gold standards (no matter the retrieval quality score, as long as it's in), the average quality score: $\frac{\sum_{j \in M_T} s_j}{|M_T|}$, where $M_T$is the set of memories from $\hat{M}_s^{ext}$and matched the given gold standard.

- False Memory Resistance $\frac{N_{miss}}{N_D}$, where $N_D$is the set of all disturbing memories, and N_{miss} is the number of disturbing memories that are successfully **ignored**. The larger this value is, the better performance the memory system will have in the face of disturbance.

- Memory Updating
  - For each D^s, a gold update standard set will be provided: $G_s^{upd} = \{(m^{old} \rightarrow m^{new})\}$
  - Normally, there will be 3 types of hallucinations in updating: neglect the new info, incorrectly modify the old info, and conflict.
  - Update Acc = $\frac{N_{correct\ updated}}{|G_s^{upd}|}$
  - Hallucination rate = $\frac{N_{wrong\ updated}}{|G_s^{upd}|}$
  - Neglect rate = $\frac{N_{neglect}}{|G_s^{upd}|}$
- Memory Question Answering
  - Flow: for each q_j, we call retrieval function, giving $\hat{R}(q_j)$, which will be further fed to the LLM to inference, resulting in $\hat{y}_j$. Then, we compare $y_j^*$and $\hat{y}_j$.
  - QA acc = $\frac{N_{correct}}{N_{total}}$,
  - QA hallucination = $\frac{N_{hallucination}}{N_{total}}$ (this is "material found, but the answer is wrong")
  - QA neglect = $\frac{N_{neglect}}{N_{total}}$(this is "material not found, so not answer")

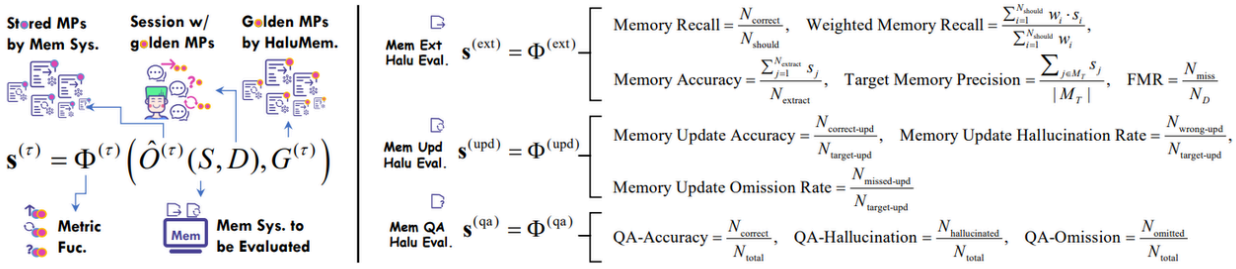

Figure 4 Hallucination evaluation process

- Work Cited
  - [Paper]
  - [Chat records]